AUTOMOBILE ENGINEERING · PART 4 OF 11

Chassis, Suspension, Steering and Brakes

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

The chassis is the skeleton that carries the engine, passengers and crash loads. Suspension keeps the tyres on the road over bumps, steering points the car, and brakes turn speed into heat. All four work through one small thing: the patch of rubber where each tyre meets the road. Stopping distance grows with the **square** of speed, which is why 120 km/h is far more dangerous than 80 km/h. For Indian roads the practical questions are ground clearance, ride comfort over potholes, and whether the car has ABS and ESC.

1. The chassis and body

Most modern cars use a **monocoque** (unibody): the floor, pillars and roof are one welded steel shell that is both body and frame. It is light, strong in a crash and cheap to mass-produce. Tough utility vehicles such as the Mahindra Thar and Scorpio-N use a separate **ladder frame**, two heavy steel rails with the body bolted on top.* It carries heavy loads and survives abuse, but is heavier and rides less smoothly. Electric cars add a new idea: the battery pack forms a flat structural floor (a 'skateboard'), which Part 6 covers.

2. Suspension: springs and dampers

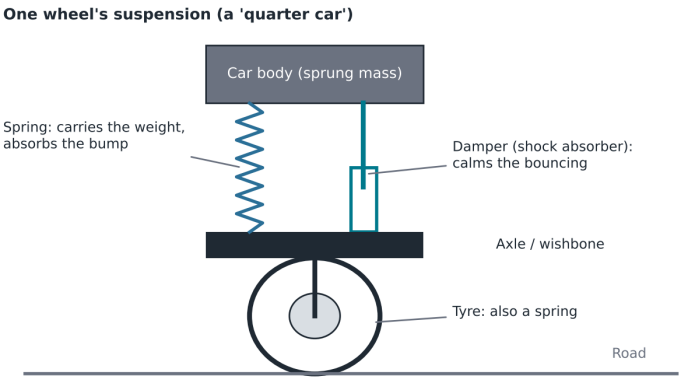


Fig 1. A spring carries the car's weight and absorbs bumps. A damper stops the spring from bouncing repeatedly. The tyre acts as a third, stiffer spring.

A spring alone would bounce up and down after every bump, so each wheel also has a **damper** (shock absorber) that converts the bouncing into heat. The art is the trade-off: soft springs ride comfortably but let the body lean in corners; stiff springs corner flat but thump over potholes. Indian buyers usually want comfort and high ground clearance, so most cars are set softer and taller than European equivalents.*

- **McPherson strut (front):** a strut combining spring and damper on a single bolt-on unit. It is compact and cheap, and used on the front of almost every mass-market car.*
- **Torsion beam (rear):** two trailing arms joined by a twisting cross-beam. Simple and cheap; common on hatchbacks and compact SUVs.*
- **Multi-link (rear):** several arms control each wheel separately, giving better ride and handling at higher cost. Typical in premium cars.

- **Solid axle with leaf or coil springs:** a rigid beam across the rear, tough for loads and off-road. Typical in ladder-frame SUVs and pickups.*

Ground clearance is the gap between the lowest part of the underside and the road. India's speed breakers and broken roads make it a headline spec for SUV buyers. Higher clearance helps on bad roads but raises the centre of gravity, which increases body roll.

3. Steering

Almost all cars use a **rack and pinion**: a small gear (pinion) on the steering column moves a toothed bar (rack) sideways, turning the front wheels. A power-assist system lightens the effort. Older cars used hydraulic pumps driven by the engine; modern ones use **electric power steering (EPS)**, a motor on the steering column. EPS saves fuel (no pump drag) and can vary the weight with speed, so the wheel is light in parking and firmer on the highway. The spec sheet's **turning radius** tells you how tight a U-turn the car can make, which matters more in a crowded city than 0–100 km/h time.

4. Brakes

Brakes work by friction: pads clamp a disc (or shoes press a drum) and convert the car's kinetic energy into heat. Kinetic energy is ½ × mass × speed², so energy, and therefore braking distance, grows with the square of speed. Add the distance covered while the driver reacts (about 1.5 seconds is a common assumption) and the numbers rise fast.

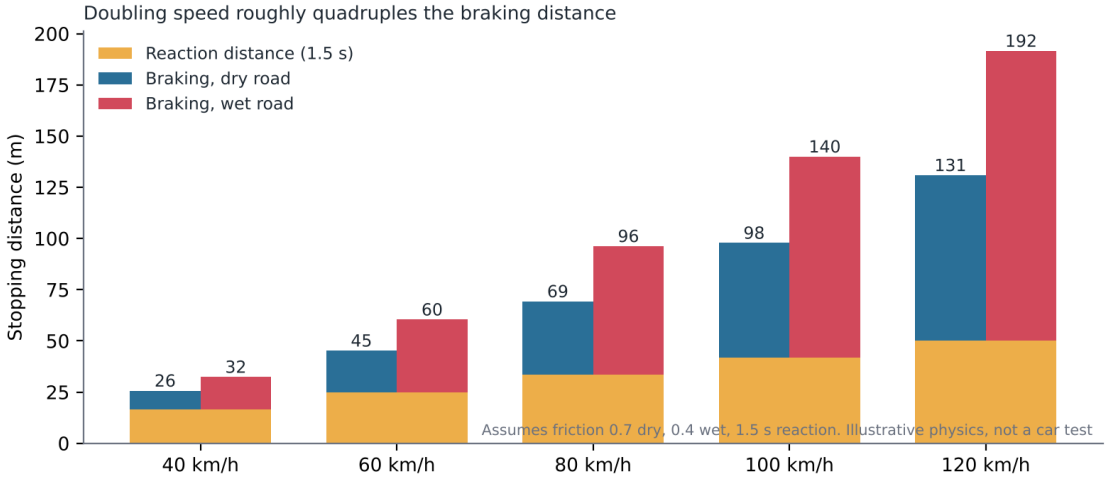


Fig 2. Total stopping distance on dry and wet roads. At 120 km/h a typical car needs roughly 130 m on a dry road and well over 150 m when wet, using simple physics with the assumptions shown.

- **Disc vs drum:** discs shed heat better and resist fade; drums are cheaper and still used on the rear wheels of many small cars.*
- **ABS (anti-lock braking):** pulses the brakes many times a second so the wheels never lock, letting you steer while braking hard. It has been compulsory on new cars in India for several years.*
- **EBD and brake assist:** distribute braking force between front and rear, and add full force when the system senses a panic stop.
- **ESC (electronic stability control):** senses a skid and brakes individual wheels to straighten the car. Many safety experts rate it as important as airbags; Part 5 covers it.
- **Regenerative braking (EVs, hybrids):** the motor works as a generator when you lift off, returning some energy to the battery. It cuts brake wear and boosts efficiency in the city.

5. Tyres: the part that touches the road

Each tyre's contact patch is only about the size of a palm, and that patch carries every force: acceleration, braking and cornering.* A size such as **215/60 R17** reads: 215 mm tread width; sidewall height equal to 60% of that width (about 129 mm); R for radial construction; 17-inch wheel. A taller sidewall (higher aspect ratio) cushions bumps; large wheels with thin sidewalls look sporty but ride harder and are costlier to replace. Tyre pressure matters for safety, comfort and mileage (Part 2), and is the cheapest thing a driver can check.

6. What it means for a buyer

- Look at ground clearance and the rear suspension type if you drive on poor roads.
- Test the ride over a speed breaker at 20–30 km/h; that tells more than a spec sheet.
- Insist on ABS and ESC, particularly on faster cars and for highway use.
- Check turning radius and steering lightness if most of your driving is in the city.

Summary

- A monocoque is a light unibody; a ladder frame is heavy and tough for utility vehicles.
- Springs carry weight; dampers control bouncing; stiffer means sportier, softer means more comfortable.
- Rack-and-pinion steering with electric assist is standard; turning radius matters in cities.
- Brakes turn kinetic energy (½ mv^2) into heat; stopping distance rises with the square of speed.
- ABS stops wheels locking; ESC stops skids; regenerative braking saves energy in EVs and hybrids.
- A tyre size code tells width, sidewall height and wheel diameter.

*Unverified or general knowledge: which cars use ladder frames, which suspension types are common, the 'palm-sized' contact patch, drum-brake use on rear wheels, the timing of the ABS mandate and the claim that Indian cars are set softer than European ones are not checked against official documents. Stopping distances in Fig 2 are calculated from simple physics with stated assumptions (friction 0.7 dry, 0.4 wet, 1.5 s reaction); real cars and tyres vary. Theory: Gillespie, Fundamentals of Vehicle Dynamics.

See also: Part 3 (transmissions). Next: Part 5, Safety engineering.